

A Gentle Ascent into Algebraic Topology: From Holes to Homotopy

Compiled from lecture notes

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1 What Algebraic Topology is About

Algebraic topology tries to answer a simple question: **How can we tell whether two spaces are truly different?** A rubber sheet can be stretched and bent without tearing, so to a topologist a coffee mug and a doughnut are “the same”

– they both have one hole. We need *invariants*: numbers, groups, or algebraic structures that stay unchanged under continuous deformations (homeomorphisms, or more generally homotopy equivalences). The three fundamental invariants we will study are:

- **Homology groups** $H_n(X)$ – detect “holes” of different dimensions.
- **Cohomology rings** $H^*(X)$ – a richer algebraic object that also remembers how holes interact.
- **Homotopy groups** $\pi_n(X)$ – classify maps from spheres into the space, extremely subtle.

We begin with the most geometric of these: homology.

2 Homology: Counting Holes with Algebra

2.1 Chains and Boundaries

To get hold of a topological space X , we approximate it by a simplicial complex – a union of points, edges, triangles, tetrahedra, etc. glued along faces. An **oriented p -simplex** is the convex hull of $p + 1$ ordered vertices $[v_0, \dots, v_p]$. Orientation means that swapping two vertices changes the sign.

The **boundary** of a simplex is the alternating sum of its faces:

$$\partial_p[v_0, \dots, v_p] = \sum_{i=0}^p (-1)^i [v_0, \dots, \widehat{v_i}, \dots, v_p],$$

where $\widehat{v_i}$ means the vertex is omitted. We extend this operator linearly to formal sums of p -simplices, the **p -chains** $C_p(K)$.

Theorem 2.1 (The boundary of a boundary is zero). $\partial_{p-1} \circ \partial_p = 0$ for all p .

Proof idea. Apply ∂_{p-1} to the above sum: each face yields a sum of $(p-2)$ -faces, and each such face appears twice with opposite signs because it can be obtained by omitting two vertices in either order.

2.2 Homology Groups

A p -chain c with $\partial_p c = 0$ is called a **cycle** (it has no boundary). A p -chain of the form $\partial_{p+1} d$ is called a **boundary**. Because $\partial^2 = 0$, every boundary is a cycle. The p -th **homology group** is

$$H_p(K) = \frac{\ker \partial_p}{\operatorname{im} \partial_{p+1}}.$$

Intuitively, homology classes are cycles that are not themselves boundaries – that is, they detect “holes” that cannot be filled in.

Example 2.1 (The circle S^1). Approximated by a triangle, the chain $a + b + c$ in the picture is a cycle but not a boundary (there is no 2-chain whose boundary is $a + b + c$). Hence $H_1(S^1) \cong \mathbb{Z}$.

Example 2.2 (The torus $T = S^1 \times S^1$). One 2-dimensional hole (the interior) and two independent 1-dimensional loops:

$$H_0(T) = \mathbb{Z}, \quad H_1(T) = \mathbb{Z} \oplus \mathbb{Z}, \quad H_2(T) = \mathbb{Z}.$$

2.3 Relative Homology and the Long Exact Sequence

Often we want to study a space modulo a subspace. For a pair (K, L) , we define **relative chains** $C_p(K, L) = C_p(K)/C_p(L)$. The boundary map descends to a map ∂_p on the quotient, and we obtain **relative homology** $H_p(K, L)$. Intuitively, chains in L are ignored.

The relationships among $H_*(L)$, $H_*(K)$ and $H_*(K, L)$ are captured by a **long exact sequence**:

$$\cdots \longrightarrow H_p(L) \xrightarrow{i_*} H_p(K) \xrightarrow{j_*} H_p(K, L) \xrightarrow{\partial_*} H_{p-1}(L) \longrightarrow \cdots$$

where $i : L \hookrightarrow K$ is inclusion, $j : K \hookrightarrow (K, L)$ is the natural map, and ∂_* is a connecting homomorphism that “pushes a relative cycle onto its boundary in L ”.

Construction and Proof of Exactness

We want to understand why this sequence is exact, i.e. $\text{Im } i_* = \ker j_*$, etc. The proof is a standard application of the **zig-zag lemma** applied to the short exact sequence of chain complexes

$$0 \longrightarrow C_*(L) \xrightarrow{i} C_*(K) \xrightarrow{j} C_*(K, L) \longrightarrow 0.$$

1. **Exactness of the chain sequence.** For each degree p , the map i is injective and j is surjective by definition, and $\text{Im } i_p = \ker j_p$ because $C_p(K, L) = C_p(K)/C_p(L)$. Hence we have a short exact sequence of chain complexes.
2. **The induced long exact sequence.** Applying homology to a short exact sequence of chain complexes always yields a long exact sequence of homology groups. The connecting homomorphism $\partial_* : H_p(K, L) \rightarrow H_{p-1}(L)$ is defined as follows:
 - Take a relative cycle $\bar{c} \in C_p(K, L)$ (so $\bar{\partial}\bar{c} = 0$). Lift $\bar{c}$ to an absolute chain $c \in C_p(K)$ via the surjection j . Then $\partial(c)$ maps to zero in $C_{p-1}(K, L)$ (because $\bar{\partial}\bar{c} = 0$), so $\partial(c) \in \ker j_{p-1} = \text{Im } i_{p-1}$. Hence there is a unique $b \in C_{p-1}(L)$ with $i_{p-1}(b) = \partial(c)$.
 - One checks that b is a cycle and its class $[b] \in H_{p-1}(L)$ is independent of the choice of lift. So we set $\partial_*[\bar{c}] = [b]$.

3. **Exactness at $H_p(K)$.** An absolute cycle c satisfies $j_*[c] = 0$ precisely when $j(c)$ is a relative boundary: $j(c) = \bar{\partial}(\bar{d})$. Lifting $\bar{d}$ to d , we get $j(c - \partial(d)) = 0$, so $c - \partial(d) \in \text{Im } i$, which means $[c]$ comes from $H_p(L)$. Thus $\ker j_* \subseteq \text{Im } i_*$. The opposite inclusion is trivial because $j \circ i = 0$ on chains, so $j_* \circ i_* = 0$.
4. **Exactness at $H_p(K, L)$ and $H_{p-1}(L)$.** Similar diagram chasing yields $\text{Im } j_* = \ker \partial_*$ and $\text{Im } \partial_* = \ker i_*$. The reader is encouraged to work through these zig-zags explicitly.

Remark 2.1. The exactness condition $\text{Im } f = \ker g$ means that a homology class dies when it can be “filled” by a class from the previous group. This algebraic pattern repeats throughout topology.

2.4 Excision and the Mayer–Vietoris Sequence

Two further powerful tools for computing homology are **excision** and **Mayer–Vietoris**.

Theorem 2.2 (Excision). If $U \subseteq A \subseteq X$ with $\bar{U} \subseteq \text{int}(A)$, then the inclusion $(X \setminus U, A \setminus U) \hookrightarrow (X, A)$ induces isomorphisms $H_p(X \setminus U, A \setminus U) \cong H_p(X, A)$ for all p .

Intuitively, cutting out a subset that does not touch the “relative boundary” does not change relative homology.

Theorem 2.3 (Mayer–Vietoris sequence). Let $X = U \cup V$ be a union of two open sets (or, for simplicial complexes, two subcomplexes). Then there is a long exact sequence

$$\cdots \longrightarrow H_p(U \cap V) \xrightarrow{\alpha} H_p(U) \oplus H_p(V) \xrightarrow{\beta} H_p(X) \xrightarrow{\Delta} H_{p-1}(U \cap V) \longrightarrow \cdots$$

Proof of Mayer–Vietoris (sketch)

Consider the chain complexes of the two subspaces and their intersection. For each p , we have a short exact sequence

$$0 \longrightarrow C_p(U \cap V) \xrightarrow{x \mapsto (x, -x)} C_p(U) \oplus C_p(V) \xrightarrow{(a,b) \mapsto a+b} C_p(U) + C_p(V) \longrightarrow 0.$$

The sum $C_p(U) + C_p(V)$ is a subcomplex of $C_p(X)$, and by a barycentric subdivision argument it computes the same homology as $C_p(X)$ (it is chain homotopy equivalent). Therefore we obtain a short exact sequence of chain complexes whose homology is the Mayer–Vietoris sequence, together with a connecting homomorphism Δ defined exactly as in the previous zig-zag lemma. The map α is induced by the two inclusions, β is the difference of the two inclusion-induced maps, and Δ is the connecting map.

2.5 Computing Homology of Spheres and Suspensions

With these sequences in hand we can compute the homology of spheres.

Theorem 2.4. The homology of the n -sphere is

$$H_p(S^n) \cong \begin{cases} \mathbb{Z} & p = 0 \text{ or } p = n, \\ \{0\} & \text{otherwise.} \end{cases}$$

Proof by induction using Mayer–Vietoris. Decompose S^n as the union of the upper hemisphere $U = S^n \setminus \{S\}$ and the lower hemisphere $V = S^n \setminus \{N\}$ (punctured at opposite poles). Each is homeomorphic to an open n -disk and thus contractible. Their intersection $U \cap V \simeq S^{n-1}$. Now apply Mayer–Vietoris:

$$\cdots \rightarrow H_p(U \cap V) \rightarrow H_p(U) \oplus H_p(V) \rightarrow H_p(S^n) \rightarrow H_{p-1}(U \cap V) \rightarrow \cdots$$

For $p \geq 2$, the middle term vanishes (contractible spaces), giving isomorphisms

$$H_p(S^n) \cong H_{p-1}(S^{n-1}).$$

For $p = 1$, we must check the connecting map. One finds $H_1(S^n) = 0$ for $n > 1$ and $H_0(S^n) \cong \mathbb{Z}$. Induction starting from S^0 (two points) yields the result.

Example 2.3 (Suspension isomorphism). If we apply Mayer–Vietoris to ΣK , the union of two cones C_+K and C_-K over K intersecting exactly in K , we obtain isomorphisms

$$H_{p+1}(\Sigma K) \cong H_p(K) \quad \text{for } p > 0.$$

This explains why the homology of a suspension shifts up by one dimension.

2.6 Euler Characteristic

If each chain group $C_p(K)$ has finite rank (say vector spaces over $\mathbb{R}$), the **Euler characteristic** is

$$\chi(K) = \sum_{p=0}^{\dim K} (-1)^p \operatorname{rk} C_p(K).$$

Thanks to the relation $\operatorname{rk} H_p = \operatorname{rk} Z_p - \operatorname{rk} B_p$ and a telescoping sum, one proves

$$\chi(K) = \sum_{p=0}^{\dim K} (-1)^p \operatorname{rk} H_p(K),$$

showing that χ is a topological invariant (even a homotopy invariant). For a sphere S^n , $\chi = 1 + (-1)^n$; for a torus, $\chi = 0$.

2.7 The Algebra of Homology: Tensor Products and Direct Sums

Homology groups are abelian groups. The **direct sum** $G \oplus H$ simply pairs elements coordinate-wise. The **tensor product** $G \otimes H$ is built from symbols $g \otimes h$ with the relation

$$(g_1 + g_2) \otimes h = g_1 \otimes h + g_2 \otimes h, \quad g \otimes (h_1 + h_2) = g \otimes h_1 + g \otimes h_2.$$

These operations appear naturally when computing the homology of product spaces. Useful algebraic facts include:

$$(G_1 \oplus G_2) \otimes H \cong (G_1 \otimes H) \oplus (G_2 \otimes H), \quad \mathbb{Z} \otimes G \cong G.$$

Theorem 2.5 (Künneth formula (simplified)). If the coefficient group is a field, then

$$H_p(X \times Y) \cong \bigoplus_{i+j=p} H_i(X) \otimes H_j(Y).$$

For integer coefficients, a torsion correction term appears.

3 Cohomology: The Dual Perspective

3.1 From Chains to Functions

If homology is built from chains, cohomology is built from functions on chains. Let $C_p(K)$ be the p -chains. Define **cochains** $C^p(K) = \text{Hom}(C_p(K), \mathbb{Z})$ (or with coefficients in a ring R). The **coboundary** operator $d_p : C^p(K) \rightarrow C^{p+1}(K)$ is the dual of ∂_{p+1} :

$$(d_p \varphi)(c) = \varphi(\partial_{p+1} c).$$

Because $d \circ d = 0$, we obtain **cohomology groups**

$$H^p(K) = \frac{\ker d_p}{\text{im } d_{p-1}}.$$

3.2 De Rham Cohomology: A Smooth View

For smooth manifolds, the most natural avatar of cohomology comes from differential forms. A **p -form** ω is an antisymmetric multilinear object that can be integrated over p -dimensional regions. The **exterior derivative** $d\omega$ raises the degree by one and satisfies $d \circ d = 0$.

A form is **closed** if $d\omega = 0$, and **exact** if $\omega = d\eta$. The **de Rham cohomology** is

$$H_{\text{dR}}^p(M) = \frac{\{\text{closed } p\text{-forms}\}}{\{\text{exact } p\text{-forms}\}}.$$

Its key property is that it is isomorphic to singular cohomology with real coefficients, so it detects the same “holes”.

Example 3.1 (Poincaré Lemma). On a contractible manifold (like a star-shaped domain in $\mathbb{R}^n$), every closed form is exact. Hence $H^p(M) \cong \{0\}$ for $p > 0$.

3.3 Wedge Product and the Cup Product

The **wedge product** of two differential forms $\omega \in \Omega^p$, $\eta \in \Omega^q$ is a $(p+q)$ -form defined by

$$(\omega \wedge \eta)(v_1, \dots, v_{p+q}) = \frac{1}{p!q!} \sum_{\sigma} \text{sgn}(\sigma) \omega(v_{\sigma(1)}, \dots) \eta(v_{\sigma(p+1)}, \dots).$$

It satisfies $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^p \omega \wedge d\eta$, so it descends to cohomology. On the cohomological level this gives the **cup product**

$$\cup : H^p(M) \times H^q(M) \longrightarrow H^{p+q}(M), \quad [\omega] \cup [\eta] = [\omega \wedge \eta].$$

This makes $H^*(M) = \bigoplus_{p \geq 0} H^p(M)$ into a graded ring. The ring structure is a powerful invariant: for example, it easily distinguishes $\mathbb{CP}^2$ from $S^2 \vee S^4$ even though their homology groups are the same.

3.4 Stokes' Theorem and Duality

The crowning jewel of de Rham theory is **Stokes' theorem**:

$$\int_{\Omega} d\omega = \int_{\partial\Omega} \omega.$$

It encapsulates the fundamental theorem of calculus and all higher-dimensional analogues. In the language of chains, it says that the evaluation pairing

$$\langle [\omega], [c] \rangle = \int_c \omega$$

between cohomology classes and homology classes is well-defined and non-degenerate over $\mathbb{R}$. This realizes $H^p(M; \mathbb{R})$ as the dual vector space of $H_p(M; \mathbb{R})$.

Remark 3.1 (Cohomology vs. Homology). Cohomology has advantages:

- A ring structure (cup product) that captures subtle intersection information.
- Natural transformation laws under smooth maps (pullback).
- Links to global analysis, e.g. the interpretation of closed forms as solutions to differential equations.

4 Homotopy Groups: Loops and Higher Spheres

Homotopy groups detect higher-dimensional winding, not just holes. The exact sequences here are just as fundamental.

4.1 The Fundamental Group

Definition 4.1. For a pointed space (X, x_0) , $\pi_1(X, x_0)$ is the set of homotopy classes of loops based at x_0 . The group multiplication is concatenation of loops.

Example 4.1. • $\pi_1(S^1) \cong \mathbb{Z}$: a loop that goes around the circle n times gives the integer n .

- $\pi_1(S^n) \cong \{0\}$ for $n \geq 2$: any loop on the sphere can be shrunk to a point.
- $\pi_1(\mathbb{RP}^2) \cong \mathbb{Z}/2\mathbb{Z}$: the generator is a loop that cannot be shrunk, but going around twice is homotopically trivial.
- $\pi_1(T^2) \cong \mathbb{Z} \oplus \mathbb{Z}$: the two independent loops around the two holes.

4.2 Higher Homotopy Groups and the Exact Sequence of a Pair

For $n \geq 2$, $\pi_n(X, x_0)$ is defined using maps $f : (I^n, \partial I^n) \rightarrow (X, x_0)$. Equivalently, maps $S^n \rightarrow X$ preserving base points. The group operation is defined coordinatewise (and is abelian for $n \geq 2$). There is a relative version $\pi_n(X, A, x_0)$ (or $\pi_n(X, A)$ for short) using maps from the n -cube that send one face to A and the remainder of the boundary to x_0 . These fit into the **long exact sequence of a pair**:

$$\cdots \longrightarrow \pi_n(A) \xrightarrow{i_*} \pi_n(X) \xrightarrow{j_*} \pi_n(X, A) \xrightarrow{\partial} \pi_{n-1}(A) \longrightarrow \cdots$$

Construction and Exactness

The maps are:

- i_* induced by inclusion $A \hookrightarrow X$,
- j_* by regarding a map $(I^n, \partial I^n) \rightarrow (X, x_0)$ as a map $(I^n, I^{n-1}, J^{n-1}) \rightarrow (X, A, x_0)$ where the face condition is satisfied,
- ∂ restricts a relative map to its face $I^{n-1} \times \{0\}$ that lands in A .

Exactness at $\pi_n(X)$. If $[f] \in \ker j_*$, then f represents a relative homotopy class that is trivial, so there is a homotopy H of maps $(I^n, \partial I^n) \rightarrow (X, x_0)$ relative to the face $I^{n-1} \times \{0\}$ (which sits in A). Actually, $j_*[f] = 0$ means that f , viewed as a relative map, is homotopic relative to the distinguished face to the constant map at x_0 . This homotopy provides a deformation of f to a map whose image lies entirely in A , hence $[f]$ comes from $\pi_n(A)$. The detailed argument uses the fact that maps $I^n \rightarrow X$ are homotopic rel ∂I^n to maps into A if they can be deformed to lie in A while fixing the boundary. Hence $\text{Im } i_* = \ker j_*$.

Exactness at $\pi_n(X, A)$. The condition $\partial[\alpha] = 0$ means that the restriction of α to the face I^{n-1} is homotopic to the constant map inside A . One can extend

this homotopy to a homotopy of the whole cube, obtaining a representative that is constant on that face, i.e. coming from an element of $\pi_n(X)$. Thus $\text{Im } j_* = \ker \partial$.

Exactness at $\pi_{n-1}(A)$. Exercise.

Remark 4.1. The homotopy exact sequence of a pair does not require a simplicial structure; it works for any pair of spaces (with some niceness assumptions). It is fundamental for relating homotopy groups of a space and a subspace.

4.3 Fibrations and the Homotopy Exact Sequence

A **fibration** $p : E \rightarrow B$ is a map satisfying the homotopy lifting property: any homotopy $H : X \times I \rightarrow B$ can be lifted to $\tilde{H} : X \times I \rightarrow E$ if an initial lift is given. Every fibre bundle (locally trivial) is a fibration. For a fibration with fibre $F = p^{-1}(b_0)$, we have the **homotopy exact sequence**:

$$\cdots \longrightarrow \pi_n(F) \longrightarrow \pi_n(E) \longrightarrow \pi_n(B) \xrightarrow{\partial} \pi_{n-1}(F) \longrightarrow \cdots$$

ending at $\pi_0(F)$, $\pi_0(E)$, $\pi_0(B)$ as sets.

Proof of Exactness (Sketch)

The key ingredient is the homotopy lifting property.

- The map $\pi_n(F) \rightarrow \pi_n(E)$ is induced by inclusion.
- The map $\pi_n(E) \rightarrow \pi_n(B)$ is p_* . Its kernel consists of classes represented by maps $f : S^n \rightarrow E$ such that $p \circ f$ is nullhomotopic. Lifting the nullhomotopy gives a homotopy of f into the fibre over b_0 , so $[f]$ comes from $\pi_n(F)$. Thus $\text{Im } i_* = \ker p_*$.
- The connecting map ∂ : given $[g] \in \pi_n(B)$, represent it by $g : (I^n, \partial I^n) \rightarrow (B, b_0)$. The constant map at e_0 lifts to a map $\tilde{g}$ defined on $I^{n-1} \times \{0\}$, and we extend it to a lift $\tilde{G}$ of g on the whole cube using the homotopy lifting property. The restriction of $\tilde{G}$ to the remaining face $I^{n-1} \times \{1\}$ lands in the fibre F , giving an element of $\pi_{n-1}(F)$. This is independent of choices and defines $\partial[g]$.
- Exactness elsewhere follows from similar lifting arguments.

Example 4.2 (Covering spaces). The universal cover $\mathbb{R} \rightarrow S^1$, $t \mapsto e^{2\pi i t}$, is a fibration with fibre $\mathbb{Z}$ (discrete). Using the exact sequence and $\pi_n(\mathbb{R}) = 0$ for all n , we deduce $\pi_n(S^1) = 0$ for $n \geq 2$ and $\pi_1(S^1) = \mathbb{Z}$.

4.4 The Hopf Fibration

A remarkable map discovered by Hopf is the fibration

$$S^1 \hookrightarrow S^3 \xrightarrow{\eta} S^2.$$

Here $S^3 = \{(z_0, z_1) \in \mathbb{C}^2 \mid |z_0|^2 + |z_1|^2 = 1\}$ and $\eta(z_0, z_1) = [z_0 : z_1] \in \mathbb{CP}^1 \cong S^2$. The long exact sequence of this fibration gives

$$\cdots \longrightarrow \pi_k(S^1) \longrightarrow \pi_k(S^3) \longrightarrow \pi_k(S^2) \xrightarrow{\partial} \pi_{k-1}(S^1) \longrightarrow \cdots$$

Since $\pi_k(S^1) = 0$ for $k \geq 2$, we obtain isomorphisms $\pi_k(S^3) \cong \pi_k(S^2)$ for $k \geq 3$. In particular, for $k = 3$, using $\pi_3(S^3) = \mathbb{Z}$ (by the Hurewicz theorem) we obtain

$$\pi_3(S^2) \cong \mathbb{Z}.$$

Thus there are non-trivial maps $S^3 \rightarrow S^2$ that cannot be deformed to a constant map, the first example of a higher homotopy group being infinite even though no homology group would predict it.

4.5 Projective Spaces and Orthogonal Groups

Further identifications connect geometry and algebra:

$$SO(3) \cong \mathbb{RP}^3 \cong S^3/\{\pm 1\}, \quad \mathbb{CP}^n \cong SU(n+1)/U(n).$$

The Hopf map is then the quotient $S^3 \rightarrow S^3/S^1 \cong S^2$, where we view S^3 as the unit quaternions and S^1 as the complex numbers of unit norm inside them.

4.6 The Hurewicz Theorem

A deep relation between homotopy and homology is given by the **Hurewicz theorem**: if X is $(n-1)$ -connected ($\pi_i(X) = 0$ for $i < n$) with $n \geq 2$, then

$$\pi_n(X) \cong H_n(X)$$

is an isomorphism. For $n = 1$, abelianization gives $H_1(X) \cong \pi_1(X)^{\text{ab}}$. This theorem explains, for instance, why the first non-vanishing homotopy and homology groups of a sphere coincide: $\pi_n(S^n) \cong H_n(S^n) \cong \mathbb{Z}$.

5 Concluding Remarks

We have covered the three pillars of basic algebraic topology:

- **Homology**: cycles mod boundaries, counts holes; easy to compute with exact sequences.
- **Cohomology**: dual theory, has a ring structure; connected to differential forms and integration via de Rham cohomology.
- **Homotopy**: maps from spheres, much richer but harder to compute; related to homology by the Hurewicz theorem.

Along the way we encountered four families of exact sequences that are the workhorses of algebraic topology:

- **Homology of a pair** and **Mayer–Vietoris** – for computing homology.
- **Homotopy of a pair** and **of a fibration** – for computing homotopy groups.

Their proofs, though elementary (diagram chasing or lifting arguments), demonstrate the power of categorical thinking. With them, one can compute invariants of a vast array of spaces.

The journey does not end here. Further topics include:

- Higher homotopy theory (e.g. $\pi_4(S^3) \cong \mathbb{Z}/2\mathbb{Z}$).
- Spectral sequences for computing homology and homotopy.
- Sheaf cohomology and K -theory.
- Topological quantum field theories and the categorification of these invariants.

As with the study of adjunctions and Yoneda, the key is to see that *every abstract invariant encodes a concrete geometric intuition*. The “holes” detected by homology, the “intersection numbers” recorded by the cup product, and the “winding” measured by homotopy are all facets of the same fundamental quest: to understand shape through algebra.